AttnWGAIN: Attention-based Wasserstein Generative Adversarial Imputation Network for IoT Multivariate Time Series

Zhaoqin Wang, *Student Member, IEEE*, Qian Ma, *Member, IEEE*, Defu Cui, Shikai Guo, *Member, IEEE*, Hui Li, *Member, IEEE*, Qiao Ning, *Member, IEEE*

***Abstract*—The missing values pervasively exist in IoT multivariate time series, which severely impairs the interpretability of the data and poses a major problem for advanced analysis and pattern recognition tasks. Existing IoT time series imputation methods typically address missing values through deletion, statistical estimation, or deep-learning-based imputation However, these methods often fail to sufficiently explore temporal information or are multi-stage. In this paper, we introduce a novel model for IoT time series imputation, called AttnWGAIN, which utilizes an attention mechanism to capture both temporal and feature correlations in the data and imputes missing values (MVs) in a generative manner. Specifically, AttnWGAIN is a GAN-based model where the generator leverages a number of diagonal masked attention blocks to reconstruct an IoT complete time series, and the discriminator attempts to distinguish between real (observed) and fake (imputed) elements to enhance imputation accuracy. Furthermore, by training the model with a joint optimization strategy, AttnWGAIN demonstrates superior performance in IoT time series imputation. We conduct extensive experiments on four real-world IoT datasets, and the results quantitatively and qualitatively show that AttnWGAIN outperforms state-of-the-art methods in IoT time series imputation tasks. The source code is available at https://github.com/wzq-come-on/AttnWGAIN.git.**

***Index Terms*—Missing data imputation; IoT multivariate time series; multi-head self-attention; data quality; deep learning.**

I. Introduction

WITH the rapid development of sensor and hardware technologies, Internet of Things (IoT) devices have become a crucial source of IoT multivariate time series data [1], whose generated multivariate time series are driving innovations and application breakthroughs in smart hardware. These devices continuously collect and transmit massive real-time time series data through sensors, which are widely applied in numerous fields such as consumer electronics [2], [3], transportation science [4], economics [5], [6], medicine [7], [8], and meteorology [9], [10]. The time series data generated by IoT are typically high-frequency, real-time, and dynamically changing, serving as the foundation for data-driven research in consumer electronics. In addition to the time dimension, each time series contains multiple variables that may be interrelated or independent. Through the analysis and mining of IoT multivariate time series, the correlations and potential patterns between variables can be revealed, thus providing powerful information support for decision-making.

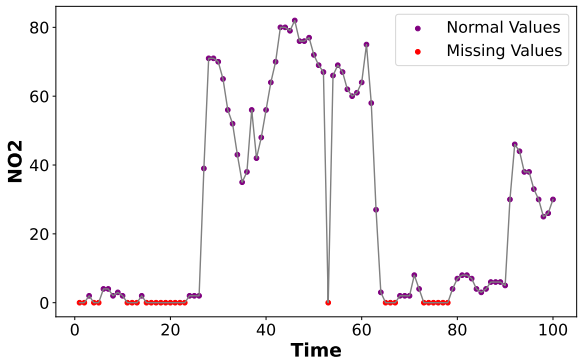


Fig. 1. MV in the real world IoT dataset (MV has a value of 0 in the figure)

However, data collected by IoT devices face more serious challenges, especially when it comes to missing value processing. IoT data often has missing values due to sensor failures, battery depletion, network delays, device drops, etc. As an example, we take a data segment from the real-world IOT dataset Air-Quality (introduced in Section V-A in detail). Fig. 1 presents the values on the 'NO2' attribute evolve with time and it can be observed that there exist many missing values (with observed values 0). Additionally, according to statistics, the proportion of missing values in UCI [1] datasets (that include many IoT time series) exceeds 40% [11], [12]. These missing values not only seriously affect the integrity and interpretability of data, but also bring challenges to data analysis and application, especially in the fields of data mining [13], [14] and pattern matching [15], [16].

Motivated by significant demand, researchers have proposed numerous missing value imputation algorithms, e.g., statistically-based mean imputation [17], [18], regression imputation [19], K-nearest neighbors imputation [20], machine

Zhaoqin Wang, Qian Ma, Shikai Guo, and Hui Li are with the College of Information Science and Technology, Dalian Maritime University, Dalian 116026, P. R. China.
E-mail: 18708975722@163.com, {maqian, shikai.guo, lihui}@dlmu.edu.cn. Qian Ma is the corresponding author.
Defu Cui is with the Air Ticket Business Department, TravelSky Mobile Technology Co., Ltd., Beijing, 100043, China.
E-mail: davecui86@gmail.com.
Qiao Ning is with the School of Information Science and Technology, Northeast Normal University, Wuxi 214122, P. R. China.
E-mail: ningq669@jiangnan.edu.cn.

[1]https://archive.ics.uci.edu/

learning-based random forest imputation [21], SVM imputation [22], [23], decision tree imputation [24], etc. However, most traditional methods are not designed for IoT multivariate time series and do not account for temporal correlations in the data. With advances in computing power and various promising applications, deep learning and neural networks (NN) have gained tremendous attention. Many deep-learning-based time-series imputation approaches, e.g., SAITS [25], GAIN [26], have been intensively proposed in recent years. Notably, the generative ability of GANs [27] provides them with inherent advantages in missing value imputation, and various GAN-based time series imputation models, e.g., GAIN [26], GAN-with-GRUI [28], E^2GAN [29], and GAMIN [30] have achieved attractive performance. Yoon et al. first introduced GANs into the MV imputation task with GAIN. In GAN-with-GRUI, Luo et al. proposed a two-stage time series imputation method that trains a GAN model in the first stage and finds the best "noise" input to make the generated data sample most similar to the original one in the second stage. Moreover, a modified Gate Recurrent Unit (GRUI) is employed in GAN to learn the temporal dependencies of observations for effective MV imputation. To reduce the time costs of the GAN-with-GRUI introduced above, the same authors proposed E^2GAN, which uses a compressing and reconstructing strategy for time series imputation. In E^2GAN, GRUI is still used for learning temporal dependencies. Seongwook Yoon et al. proposed GAMIN, a generative adversarial multiple imputation network for highly missing data. Although all the aforementioned GAN-based methods can achieve reasonable imputed results and even state-of-the-art performance, they are mostly based on recurrent neural networks (RNN), which struggle to model long-term temporal dynamics and suffer from slow speed and memory constraints. Furthermore, GANs are prone to mode collapse, which is particularly problematic for time series with diverse patterns over time. In contrast, Transformer-based models leveraging self-attention mechanisms excel in modeling long-term temporal dynamics. By capturing global dependencies through parallel computation, self-attention (as adopted in SAITS [25]) efficiently models correlations across distant time steps. The integration of self-attention's long-range dependency modeling with GAN's generative capacity overcomes the limitations of standalone approaches: attention models gain more realistic data distribution fidelity from GANs, while GANs leverage attention to break through the bottleneck in capturing complex temporal patterns, ultimately forming a complementarity between temporal dynamics modeling and data distribution fitting. Additionally, GAN-attention combinations have been explored in computer vision (e.g., ActFormer [31] for human motion generation) and natural language processing (e.g., maGAN [32] for Citation Intent Classification), achieving promising results.

Before the rise of self-supervised learning, time series imputation primarily relied on non-self-supervised methods, such as statistical imputation [17], [18] and deep learning models based on Recurrent Neural Networks (RNNs) [33]. However, these methods face critical limitations: statistical approaches overlook temporal correlations, while RNN-based models struggle to model long-term dependencies and suffer from slow inference speeds [34]. Even generative models like GANs, which show promise in imputation tasks, often suffer from mode collapse and require substantial labeled data for training [26].

Recent studies have demonstrated that self-supervised learning brings a paradigm shift to sequence imputation. For example, Zhang et al. [35] proposed a graph contrastive learning framework GCL4SR, introducing a self-supervised framework that leverages acceleration signals to design masking strategies and diffusion processes. By forcing the model to reconstruct time series representations from partial observations, it enhances the robustness of sequence representations. Yang et al. [36] proposed a debiased contrastive learning framework DCRec, which employs contrastive learning—a self-supervised paradigm—to integrate graph structures and disentangle conformity behaviors, thereby improving representation learning for sequential recommendation. Inspired by these works, this study is dedicated to integrating self-supervised principles into generative models for IoT time series imputation.

To address the issues mentioned above, we propose a novel *Attention-based Wasserstein Generative Adversarial Imputation Network* (AttnWGAIN) for IoT multivariate time series. The model employs self-supervised learning techniques, enabling it to learn robust temporal representations without requiring manual annotations. The proposed AttnWGAIN consists of a generator and a discriminator. The generator aims to impute missing values in IoT time series by generating complete data samples that closely resemble real ones, while the discriminator strives to differentiate between real (observed) and fake (imputed) elements. To enhance imputation accuracy, we integrate the multi-head attention mechanism into the generator, leveraging its strengths in handling long sequences and enabling parallel processing of input data. Additionally, to mitigate the problem of mode collapse,we employ the Wasserstein GAN (WGAN), which provides a smoother gradient and helps stabilize the training process. To the best of our knowledge, no existing work combines GANs with attention mechanisms to tackle the problem of IoT time series imputation. The main contributions are summarized below.

- We propose a novel generative IoT time series imputation model, AttnWGAIN. By updating the generator and discriminator with a joint-optimization training strategy, the well-trained AttnWGAIN can learn the distribution of the entire dataset and outperform state-of-the-art (SOTA) methods.
- To effectively capture the long-term temporal dependencies and feature correlations explicitly, we employ a transformer encoder consisting of multiple diagonally-masked multi-head attention (DiagMaskedMHA) blocks to implement the generator.
- We model a time lag matrix to reflect the impact of the past observed values on the current imputation. By including the time lag matrix as part of AttnWGAIN's input, the model can effectively capture changing temporal correlations among observations.
- We conduct extensive experiments over four real-world

IoT datasets to quantitatively and qualitatively evaluate the proposed AttnWGAIN. The experimental results show that AttnWGAIN achieves new state-of-the-art imputation accuracy.

The rest of the article is as follows. Section II and Section III introduce related works and problem formulation. Section IV gives complete details of the proposed AttnWGAIN. Section V presents the experimental results. Finally, the conclusion and future work is provided in Section VI.

II. Related Works

Several missing value imputation methods have been proposed in recent years. However, due to space constraints, we restrict our discussion to the works most directly related to time-series imputation. Inspired by SAITS, we categorize the related studies into five groups.

RNN-based: Che et al. [37] proposed a new gated recurrent unit (GRU)-based variant, GRU-D, for dealing with MVs in time series classification. Moreover, this work first introduced the concept of time decay which was later adopted in subsequent works such as [28], [29], [34], [38]. Both M-RNN [38] and BRITS [34] utilize the hidden states of bidirectional RNNs to estimate missing values. "The difference is that while M-RNN assumes missing values (MVs) are fixed constants, BRITS models them as variables within the RNN network.

GNN-based: The GNN-based model treats the time series as a sequence of graphs and reconstructs the MVs using learned node embeddings. "Cini et al. introduced GRIN [39], an innovative graph-based recurrent model for Multilayer Time Series Imputation (MTSI). GRIN utilizes a bi-directional graph recurrent neural network to efficiently exploit both temporal dynamics and spatial similarity, thus significantly improving the imputation accuracy. In addition, Marisca et al. proposed SPIN [40], which integrates a unique sparse spatio-temporal attention mechanism into the GNN framework.

GAN-based: Models [26], [28], [33], [38] are also based on RNNs. However, they are categorized separately as GAN-based models due to their utilization of the generative adversarial network (GAN) structure. GAIN (Generative Adversarial Estimation Network) proposed by Yoon et al [26] handles different types of missing values in tabular data. Luo et al. [28] introduced a Gated Recurrent Unit (GRUI) unit derived from GRU to capture the temporal irregularities of time series. Additionally, Luo et al. [29] proposed another imputation method, namely E^2GAN, which is built upon the foundation laid in [28]. Yoon et al. proposed the GAMIN [30] for confidence-based prediction. They implemented a recursive architecture with GAMIN and trained it with a novel loss function based on confidence, which effectively fills in highly missing data. Liu et al. proposed a non-autoregressive model NAOMI [33] whose performance is improved by employing adversarial training techniques.

VAE-based: VEA model shows great potential for MV imputation as it provides a probabilistic manner to explore the correlations amongst data with mathematical theory. Many existing MV imputation methods, e.g., MIWAE [41], GP-VAE [42], HI-VAE [43], L-VAE [44], SGP-VAE [45] and MIVAE [46], are developed based on the VAE model. In these works, the Gaussian process (GP) prior is employed in the latent space for time series or exploring the temporal correlations of data.

Diffusion-based: Diffusion models are capable of filling MVs by approximating the posterior distribution of data conditioned on observed values. The CSDI [47] is a time series imputation method that employs the conditional score-based diffusion model, where the temporal and feature correlations are captured through a two-dimensional attention mechanism. TabCSDI [48] modifies the CSDI introduced above for tabular data, where how to handle the categorical variables is discussed deeply. PriSTI [49] adopts diffusion model to fill MVs based on the observed values and underlying spatiotemporal dependencies of data. JML Alcaraz et al. [50] proposed an SSSD model that is based on the conditional diffusion model and structured state space model to capture the long-term dependencies in time series for MV imputation. Compared with the existing autoregressive interpolation models, the latest diffusion-based imputation works show a better performance in MV imputation.

Attention-based: Since the Transformer model was introduced by Vaswani et al. [51], self-attention techniques have been widely adopted for modeling sequential data, with a particular focus on time series. Ma et al. [52] proposed CDSA, a model for estimating geo-tagged spatiotemporal data through co-learning of time, location, and measurements. Bansal et al. [53] proposed DeepMVI that integrates transformer and convolutional techniques to extract long-term seasonality, granular localization, and cross-dimensional embeddings for imputing IoT multivariate time series. Du et al. [25] proposed SAITS, a model that utilizes two Transformer encoders to capture both time-based dependencies and feature correlations at each time step.

Even though numerous time series imputation methods have been proposed, different solutions address the missing data issue from various perspectives and are suited to different situations. Therefore, no single method is universally the best, and each class of solutions has certain limitations, as summarized below.

RNN-based imputation models, although good at capturing sequential information, they are affected by the limitations of the recurrent network structure. They tend to take a long time and have limited memory resources, which severely limits their ability to capture long-term dependencies when processing time series data, leading to complex errors in time series analysis. In addition, most RNN models have autoregressive properties, which in the field of time series analysis [54] can cause so-called compound error problems. GNN-based models approaches provide a deeper perspective for understanding and analyzing spatio-temporal dynamics, however, this is often accompanied by a significant increase in computational complexity. This makes their use challenging in the context of large-scale IoT multivariate time series. GAN-based, VAE-based, and diffusion-based models are generative models that have demonstrated effectiveness in both deterministic imputation [8]–[10] and probabilistic imputation [55]. However, GAN models often suffer from non-convergence and

TABLE I
A SEGMENT EXAMPLE OF INCOMPLETE TIME SERIES

Time/Dimension	GCS	HR	NIDiasABP	NIMAP	NISysABP	Temp	Urine	Weight
0:02 ($\mathbf{x}_{t_1}$)	7	100	84	100.3	133	37.8	100	56.7
0:47 ($\mathbf{x}_{t_2}$)	?	78	83	99	131	?	?	56.7
1:17 ($\mathbf{x}_{t_3}$)	?	96	89	105	137	?	60	56.7
2:17 ($\mathbf{x}_{t_4}$)	8	81	89	106.7	142	37.2	75	56.7
2:47 ($\mathbf{x}_{t_5}$)	9	101	82	97.67	129	?	?	56.7
3:17 ($\mathbf{x}_{t_6}$)	8	86	75	92.67	125	39	30	56.7
4:17 ($\mathbf{x}_{t_7}$)	8	87	72	101.3	134	37.9	125	56.7
5:17 ($\mathbf{x}_{t_8}$)	7	103	78	106	148	38.6	100	56.7
5:47 ($\mathbf{x}_{t_9}$)	?	86	89	98.67	140	39.2	120	56.7
...	...	...	...	...	...	...	...	...

mode collapse [56], necessitating a carefully designed training process for multivariate imputation. VAE-based models are adept at explicitly modeling probabilities but can be limited by generative power, potentially reducing their performance in capturing complex data variability. Besides the considerable computational complexity, diffusion-based models suffer from problems related to the coherence of the boundaries between missing and observed parts [57]. Attention-based models usually outperform RNN-based models due to their advantages in handling remote dependencies and parallel processing capabilities. However, they increase the computational cost when dealing with long sequences, which can lead to slower training and inference.

III. Problem Statement

In this section, we start by introducing some notation and definitions used in this paper and state our research goal.

Definition 1: An **IoT multivariate time series**, denoted by $\mathbf{X} = (\mathbf{x}_{t_1}, \mathbf{x}_{t_2}, \cdots, \mathbf{x}_{t_n}) \in \mathbb{R}^{n\times d}$, is a sequence of observations arriving at time points $\mathbf{T} = (t_1, t_2, \cdots, t_n)$, where $\mathbf{x}_{t_i}$ is a d-dimensional observation of $\mathbf{X}$, and $x_{t_i}^j$ is the jth variable of $\mathbf{x}_{t_i}$.

Notable that if $\exists x_{t_i} \in \mathbf{X}$ where the observed value of $\mathbf{x}_{t_i}^j \in \mathbf{x}_{t_i}$ is missing, then $\mathbf{x}_{t_i}$ is termed as an incomplete observation and $\mathbf{X}$ is termed as an incomplete time series.

Definition 2: The **mask matrix** $\mathbf{M} \in \mathbb{R}^{d\times n}$ is used to indicate the missingness of the IoT multivariate time series $\mathbf{X}$. Each element $m_{t_i}^j$ of $\mathbf{M}$ denotes the missingness of the corresponding value $x_{t_i}^j$ and satisfies that $m_{t_i}^j = 1$ if $x_{t_i}^j$ is observed, otherwise $m_{t_i}^j = 0$.

Example 1: Table I shows a segment of time series sampled from the dataset PhysioNet-2012[2]. The dataset contains 12,000 multivariate clinical time series samples from intensive care unit (ICU) patients. Each observation (each row in the table) has eight attributes that are concentrations for Glasgow Coma Score (GCS), Heart rate (HR), Non-invasive diastolic arterial blood pressure (NIDiasABP), Non-invasive mean arterial blood pressure (NIMAP), Non-invasive systolic arterial blood pressure (NISysABP), Temperature (Temp), Urine output(Urine) and Weight. For illustration, we use '?' to indicate the MVs. As shown, the incomplete observations are $\{\mathbf{x}_{t_2}, \mathbf{x}_{t_3}, \mathbf{x}_{t_5}, \mathbf{x}_{t_9}\}$ (i.e., the values on some attributes of these observations are missing) and the complete observations are $\{\mathbf{x}_{t_1}, \mathbf{x}_{t_4}, \mathbf{x}_{t_6}, \mathbf{x}_{t_7}, \mathbf{x}_{t_8}\}$. Since incomplete observations exist, the time series is definitely incomplete.

Note that in this paper, we assume the missing mechanism is MCAR (missing completely at random), i.e., any value $x_{t_i}^j$ $(t_i \in \mathbf{T}, 1 \le j \le d)$ may be missing. On the other hand, due to the presence of missing values (MVs), the time lag between two consecutive valid observations is constantly changing. In a complete and regular time series, the time intervals between data points are typically fixed. However, when MVs occur, this regularity is disrupted. Taking the Temp dimension in Table I as an example, if there is a missing value at 0:47, the time lag between the valid reading at 0:02 and the next valid reading at 1:17 becomes 60 minutes, instead of the usual 30-minute interval. This change in time lag is not merely a numerical difference; it indicates that the strength of temporal correlations is also dynamically variable, which must be considered during missing value (MV) imputation, and it is of great significance for understanding the underlying patterns of the time series. Therefore, we model the time lag information as a time lag matrix $\mathbf{\Delta} \in \mathbb{R}^{d\times n}$ and incorporate it as a part of the imputation model's input. Specifically, we use the calculation algorithm from [29] to derive the time lag matrix $\mathbf{\Delta}$, with the formulation for each specific time lag $\delta_{t_i}^j \in \mathbf{\Delta}$ shown below.

$$\delta_{t_i}^j = \begin{cases} t_i - t_{i-1} & \text{if } m_{t_{i-1}}^j = 1 \,\&\, i > 0 \\ \delta_{t_{i-1}}^j + t_i - t_{i-1} & \text{if } m_{t_{i-1}}^j = 0 \,\&\, i > 0 \\ 0 & \text{if } i == 0 \end{cases} \quad (1)$$

Example 2: For the incomplete time series in Example 1, the corresponding mask matrix $\mathbf{M}$ and time lag matrix $\mathbf{\Delta}$ are shown in Fig. 2.

Problem Formulation: Given an incomplete time series $\mathbf{X}$, we aim to obtain an approximate complete time series $\widetilde{\mathbf{X}}$ where the MVs in $\mathbf{X}$ are replaced by plausible estimations derived by an MV imputation model $\mathcal{M}(\mathbf{X})$ (namely AttnWGAIN in this paper).

IV. Methodology

In this section, we first give an overview of the proposed AttnWGAIN. Then we show the details of each component of the model.

A. AttnWGAIN-an Overview

As illustrated in Figure 3, similar to traditional GANs, the proposed AttnWGAIN consists of a generator (G) and a

[2]https://www.physionet.org/files/challenge-2012/1.0.0

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 45 & 45 & 45 & 45 & 45 & 45 & 45 & 45 \\ 75 & 30 & 30 & 30 & 30 & 75 & 75 & 30 \\ 135 & 60 & 60 & 60 & 60 & 135 & 60 & 60 \\ 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 \\ 30 & 30 & 30 & 30 & 30 & 60 & 60 & 30 \\ 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 \\ 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 \\ 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 \end{bmatrix}$$

(a) mask matrix (b) time lag matrix

Fig. 2. The mask matrix and time lag matrix corresponding to the time series in Table I

discriminator (D). In traditional GANs, the generator maps noise vectors to data samples that follow the distribution of the training set, while the discriminator distinguishes between fake (generated) data and real (training) data. In our approach, the generator leverages the transformer encoder and attention mechanism. By exploring the latent relationships of features and temporal correlations amongst observations, the generator uses stacked encoding blocks (each consisting of a self-attention sub-block and a feed-forward sub-block) to reconstruct a complete time series $\widehat{\mathbf{X}}$. Meanwhile, the discriminator attempts to discern which components are real (observed) and which are fake (imputed) in $\widehat{\mathbf{X}}$, rather than determining whether the entire time series is real or fake. By training the generator and discriminator with a joint-optimization training strategy, the ultimate aim of AttnWGAIN is to fill the MVs in $\mathbf{X}$ by making $\widehat{\mathbf{X}}$ as close as possible to $\mathbf{X}$. Next, we will introduce the detailed network architecture of the generator and discriminator, and present the details of the model training strategy.

B. Generator Network

1) Network Input: Given an incomplete time series $\mathbf{X}$, the generator G takes the noisy time series $\overline{\mathbf{X}}$, the mask matrix $\mathbf{M}$ and the time lag matrix Δ derived by $\mathbf{X}$ as input. Specifically, the mask matrix $\mathbf{M}$ and the time lag matrix Δ can be computed based on their definitions once $\mathbf{X}$ is given. For the noisy time series $\overline{\mathbf{X}}$, we first generate a random noise matrix $\mathbf{Z} \in \mathbb{R}^{n \times d}$ (independent of all other variables) and let $\overline{\mathbf{X}} = \mathbf{M} \odot \mathbf{X} + (1 - \mathbf{M}) \odot \mathbf{Z}$. Note that rather than initially replacing the MVs in $\mathbf{X}$ with 0 (adopted as the default value in most of existing works), we initialize each MV $x_{t_i}^j$ ($x_{t_i}^j \in \mathbf{X}$ and $m_{t_i}^j = 0$) with a random noise $z_{t_i}^j \sim \mathcal{N}(0, 1)$, to help the generator learn more possible data distributions.

Afterwards, we concatenate $\overline{\mathbf{X}}$, $\mathbf{M}$, Δ and feed the concatenation into a linear layer to get the final embedding of the input shown below.

$$\mathbf{X}_G = LN(\text{Concat}(\overline{\mathbf{X}}, \Delta, \mathbf{M})) \tag{2}$$

2) Network Architecture: As shown in Figure 3, the generator consists of a positional encoding layer, N stacked encoding blocks, and a linear layer, where the architecture is close to the Transformer encoder.

Positional encoding. As each observation in a time series simultaneously flows through the Transformer's encoder stack, the self-attention mechanism lacks awareness of the position or order of each observation. To address this, a positional encoding layer is introduced to provide the model with information about the position of observed values within the time series sequence. While time lag information is incorporated during data preprocessing to enable the generator to learn the temporal aspects of the input time series, it primarily influences the impact of past observed values on current imputation rather than the order of the input. The inclusion of relative position encoding remains beneficial for capturing temporal dependencies and feature correlations between observations.

Specifically, for the i-th observation $\mathbf{x}_{t_i}$ in the time series $\mathbf{X}$, the computation of its position encoding is the same as that in the Transformer shown below.

$$\text{PE}(\mathbf{x}_{t_i}) = \begin{cases} sin(w_k \cdot i), & \text{if } i == 2k \\ cos(w_k \cdot i), & \text{if } i == 2k+1 \end{cases} \tag{3}$$

where $w_k = \frac{1}{10000^{\frac{2k}{d_m}}}$, and d_m denotes the dimension of the derived position encoding (taken as 256 in this paper).

Stacked encoding blocks. As previously mentioned, there are N stacked encoding blocks in the generator. Each encoding block is primarily composed of a diagonally masked multi-head attention sub-block and a feed-forward sub-block, and the composition of these sub-blocks will be detailed below.

In the domain of time series analysis, the traditional Multi-Head Attention (MHA), a groundbreaking technique introduced by Vaswani et al. [51] in 2017, has demonstrated remarkable advantages in time series imputation tasks. From the perspective of sequence dependency modeling, MHA excels in capturing both long-term and short-term dependencies across different scales within time series data by computing multiple attention heads in parallel. This capability enables the effective exploration of complex correlations among data points. In terms of feature fusion, the weighted combination of outputs from individual attention heads allows the model to extract features from diverse perspectives, thereby enhancing its ability to represent temporal patterns. Moreover, the self-attention mechanism in MHA permits each time step to interact with all other time steps in the sequence, circumventing the vanishing gradient problem that plagues recurrent neural networks when processing long sequences, thus improving the accuracy of imputation.

However, traditional MHA exhibits limitations when handling time series data with strong temporal order constraints. Specifically, it may inadvertently incorporate information from future time steps, leading to logical inconsistencies in the imputation results. Inspired by SAITS [25], we adopt the Diagonally Masked Multi-Head Attention (DiagMaskedMHA) to replace the conventional MHA. In contrast to traditional MHA, DiagMaskedMHA introduces a diagonal masking mechanism that strictly restricts each time step to attending only to preceding or concurrent time steps during the computation of attention weights. This fundamental innovation eliminates the issue of future information leakage, aligns seamlessly with the causal nature of time series data, and enables more precise modeling of dynamic temporal evolution processes.

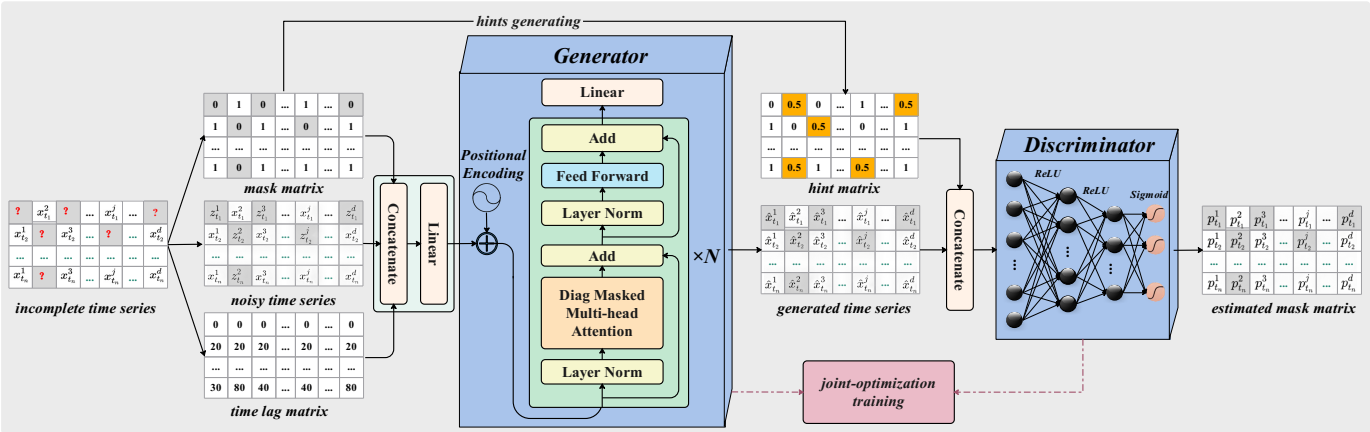


Fig. 3. Architectural Overview of the AttnWGAIN Model. The framework consists of two primary components: a generator G and a discriminator D, which operate in a tandem manner. The G comprises N stacked encoding blocks (where each block consists of a diagonally masked multi-head attention sub-block and a feed-forward sub-block) to generate a complete time series. The D is designed to differentiate between real (observed) and fake (imputed) values within the generated time series to help G achieve high imputation accuracy.

Consequently, it significantly enhances the performance and reliability of AttnWGAIN in time series imputation tasks. The mathematical formulation of DiagMaskedMHA is presented as follows:

$$\begin{aligned}&\text{DiagMaskedMHA}(Q,K,V)\\&\qquad = Concat(head_1, head_2, \cdots, head_h)W^O\end{aligned} \tag{4}$$

where Q, K, and V are the *query vector*, *key vector*, and *value vector* with d_k, d_k, and d_v dimensions, respectively. Moreover, the The parameter h is the number of heads and W^O is an output transformation matrix. The $head_i$ is the output of the i-th attention head and is defined as

$$\begin{aligned}head_i &= \text{DiagMaskedAttn}(Q_i, K_i, V_i)\\&= \text{Softmax}\left(\text{DiagMask}\left(\frac{Q_iK_i^T}{\sqrt{d_k}}\right)\right)V_i\end{aligned} \tag{5}$$

where $Q_i = QW_i^Q$, $K_i = KW_i^K$, $V_i = VW_i^V$, and $\left(W_i^Q, W_i^K, W_i^V\right)$ are the weight matrices corresponding to the (Q, K, V) of the i-th head respectively. Different from the traditional MHA where the i-th head $head_i = \text{Softmax}\left(\frac{Q_iK_i^T}{\sqrt{d_k}}\right)V$, the DiagMaskedMHA sets the diagonal value of the $\left(\frac{Q_iK_i^T}{\sqrt{d_k}}\right)$ as $-\infty$ so the diagonal attention weights approach 0 after the softmax function. Specifically, the DiagMask function is formulated as

$$\text{DiagMask}\left(\frac{QK^T}{\sqrt{d_k}}\right) = \left(\frac{QK^T}{\sqrt{d_k}}\right)\cdot A_m \tag{6}$$

$$A_m(i,j) = \begin{cases}-\infty & \text{if } i = j\\ 1 & \text{if } i \neq j\end{cases} \tag{7}$$

In the context of time series imputation, the DiagMask operation employs a specialized masking strategy to enforce that the model relies solely on observed data points when estimating missing values (MVs). This approach effectively eliminates potential self-referential loops in the imputation process. By design, the Diagonally Masked Multi-Head Attention (DiagMaskedMHA) mechanism enabled by DiagMask can simultaneously capture both temporal dependencies and feature correlations within a single attention computation, thereby significantly enhancing the model's capability to identify complex patterns in time series data. However, the internal structure of the multi-head attention layer is primarily composed of linear transformations, which are less effective at learning complex patterns compared to non-linear transformations. Consequently, while the output of the multi-head attention layer leverages the attention mechanism to construct a new representation of each data point, this representation may not be as robust as possible. The reconstructed representation of the data can be significantly enhanced using an activation function. Therefore, following the DiagMaskedMHA sub-block, a feed-forward sub-block is introduced.

The feed-forward sub-block consists of two fully connected layers, where each operates as a position-wise function. This design ensures that each input vector is processed independently of other input vectors, enabling the model to extract features specific to each position without being affected by other vectors. Between these two layers, a non-linear transformation is applied using an activation function such as ReLU. Given an input vector $\mathbf{x}_f \in \mathbb{R}^d$ for the current feed-forward sub-block, the feed-forward network FFN$(\cdot)$ can be expressed as:

$$\text{FFN}(\mathbf{x}_f) = \text{ReLU}(\mathbf{x}_f \cdot \mathbf{W}_1 + \mathbf{b}_1)\cdot \mathbf{W}_2 + \mathbf{b}_2 \tag{8}$$

where $\mathbf{W}_1$, $\mathbf{b}_1$, $\mathbf{W}_2$, and $\mathbf{b}_2$ are the learnable parameters. As stated in Equ (8), the feed-forward sub-block plays a vital role in learning the output distribution and thereby benefits the time series imputation.

Besides the two sub-blocks introduced above, the layer normalization and residual connection are also the key components of the Transformer. For any d_L dimensional input vector $\mathbf{x}_L$, the layer normalization is computed as

$$\text{LayerNorm}(\mathbf{x}_L) = \frac{\mathbf{x}_L - \mu}{\sigma}\cdot \mathbf{W}_L + \mathbf{b}_L \tag{9}$$

where μ and σ are the mean and standard deviation of the $\mathbf{x}_L$, i.e., $\mu = \frac{1}{d_L}\sum_{k=1}^{d_L} x_L^k$ and $\sigma = \sqrt{\frac{1}{d_L}\sum_{k=1}^{d_L}(x_L^k - \mu)^2}$.

$\mathbf{W}_L$ and $\mathbf{b}_L$ are the parameters. The residual connection (corresponding to the 'Add' layer in Figure 3) is mainly used to optimize the model training. The information from the previous layer is copied twice. One copy is directly passed to the next layer without any difference, and another copy is transformed through the current net. Finally, the results of these two parts are then added together as the output of the next layer. Based on the residual connection, the information from the previous layer can be retained at least, which effectively improves the model performance.

Moreover, as shown in Figure 3, both sub-blocks are preceded by a layer normalization unit and followed by a residual connection, forming what is known as the Pre-LN structure [58]. In contrast, the original Transformer design incorporates layer normalization between the residual connections, a configuration commonly referred to as Post-LN (Post-Layer Normalization) [59]. The key difference is that Pre-LN integrates layer normalization within the residual connection itself. According to [58], the Pre-LN configuration offers advantages in terms of gradient behavior. Specifically, it helps maintain well-behaved gradients, avoiding issues such as exploding or vanishing gradients, both theoretically and empirically. This stability can be crucial for training deep models effectively. Finally, the generator in this model processes the input $\mathbf{X}_G$ (as introduced in Section IV-B1) with N stacked encoding blocks. The output, denoted as $\widehat{\mathbf{X}}$, is produced through a linear layer. The data processing flow can be summarized as follows:

$$\mathbf{X}_{\text{MHA}}^{i} = \text{Add}\left(\mathbf{X}_{e}^{i-1}, \text{DiagMaskedMHA}\left(\text{LN}\left(\mathbf{X}_{e}^{i-1}\right), \text{LN}\left(\mathbf{X}_{e}^{i-1}\right), \text{LN}\left(\mathbf{X}_{e}^{i-1}\right)\right)\right) \tag{10}$$

$$\mathbf{X}_{\text{FFN}}^{i} = \text{Add}\left(\mathbf{X}_{\text{MHA}}^{i}, \text{FFN}\left(\text{LN}\left(\mathbf{X}_{\text{MHA}}^{i}\right)\right)\right) \tag{11}$$

$$\widehat{\mathbf{X}} = \text{ReLU}\left(\left\{\mathbf{X}_{\text{FFN}}^{i}\right\}_{i=2}^{N} \cdot \mathbf{W}_G + \mathbf{b}_G\right) \tag{12}$$

where $\mathbf{X}_e^i$ $(1 \leq i \leq N)$ denotes the output of the i-th encoding block, and $\mathbf{X}_e^1 = \mathbf{X}_G$ at the first time point t_1.

C. Discriminator Network

In the GAN framework, we introduce a discriminator (D) to act as an adversary for training the generator (G). The primary task of the discriminator in our AttnWGAIN model is to differentiate between imputed (fake) and observed (real) values, which amounts to predicting the mask matrix $\mathbf{M}$. Typically, both the generator and discriminator have the same network structure [60]. However, in this study, we simplify the network by implementing the discriminator with two fully connected layers. This simpler network structure achieves performance comparable to that of a discriminator utilizing an attention mechanism, as employed in the generator (see Section V-E). Additionally, to facilitate the convergence of the discriminator and ensure the generator reaches an optimal distribution concerning the discriminator, we incorporate the hint mechanism proposed in GAIN [26].

The hint mechanism involves using a *hint matrix* $\mathbf{H}$ as an additional input to the discriminator. The hint matrix is a random mask of $\mathbf{M}$, where the true mask information for a specified proportion of $\mathbf{M}$ is provided to $\mathbf{H}$. We adopt the same strategy as GAIN to generate $\mathbf{H}$ for the time series $\mathbf{X}$. The discriminator D then takes the concatenation of the generated time series $\widehat{\mathbf{X}}$ and the $\mathbf{H}$ as input. Following data processing steps outlined in Equations (13) and (14), D outputs a probability matrix $\mathbf{P} \in \mathbb{R}^{n \times d}$, where each component $p_{t_i}^j$ corresponds to the $m_{t_i}^j$ and represents the probability that $x_{t_i}^j$ is observed.

$$\mathbf{P}_1 = \text{ReLU}\left(\text{Concat}\left(\widehat{\mathbf{X}}, \mathbf{H}\right) \cdot \mathbf{W}_D^1 + \mathbf{b}_D^1\right) \tag{13}$$

$$\mathbf{P} = \text{Sigmoid}\left(\text{ReLU}\left(\mathbf{P}_1 \cdot \mathbf{W}_D^2 + \mathbf{b}_D^2\right) \cdot \mathbf{W}_D^3 + \mathbf{b}_D^3\right) \tag{14}$$

D. Model Training

A traditional GAN is through a min-max game, where the discriminator D estimates the probability that a sample originates from the training data rather than the generator G, while G seeks to maximize the likelihood of D making an error. However, traditional GANs often encounter the mode collapse problem [60]. To address this issue, we employ the Wasserstein GAN [61] with the following formulations.

$$\begin{aligned} &\underset{G}{min}\ \underset{D}{max}\ \mathcal{L}(D, G) \\ &\mathcal{L}_D = \mathbb{E}_{x \sim p_x}[D(x)] - \mathbb{E}_{z \sim q_z}[D(G(z))] \\ &\mathcal{L}_G = -\mathbb{E}_{z \sim q_z}[D(G(z))] \end{aligned} \tag{15}$$

where q_z and p_x represent the data distributions of the low-dimensional random variable z and the real data sample x. Unlike the traditional GAN, where D aims to determine whether a given data sample is fake or real, the D in our AttnWGAIN discriminates the authenticity of each element within a given time series. We train D to assign high probabilities to true observed values and low probabilities to imputed values. The specific loss function for the discriminator D in AttnWGAIN is presented below.

$$\mathcal{L}_D = \frac{1}{nd} \sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left[m_{t_i}^j \cdot p_{t_i}^j - (1 - m_{t_i}^j) \cdot p_{t_i}^j\right] \tag{16}$$

For generator G, its purpose is to make the generated time series $\widehat{\mathbf{X}}$ as similar as possible to the real time series $\mathbf{X}$ in order to accurately impute the missing values. To achieve this, we supplement the original discriminative loss (as shown in Equation (16)) with a *reconstruction loss* in the loss function of G. The reconstruction loss is defined by the square error between the original time series $\mathbf{X}$ and the generated time series $\widehat{\mathbf{X}}$. It is important to note that the ground truths of the missing components in the original $\mathbf{X}$ are unknown, which means we can only calculate the reconstruction error of the observed components. As illustrated in SAITS [25], the non-autoregressive attention-based model can distinguish the observed part and minimize the reconstruction error only for those observed values in $\mathbf{X}$. Consequently, the generated value for the unobserved (missing) parts tend to have a high reconstruction error, leading to unsatisfactory imputation results. To address this, we manually mask some observed values for joint optimization, similar to the approach in SAITS.. Specifically,

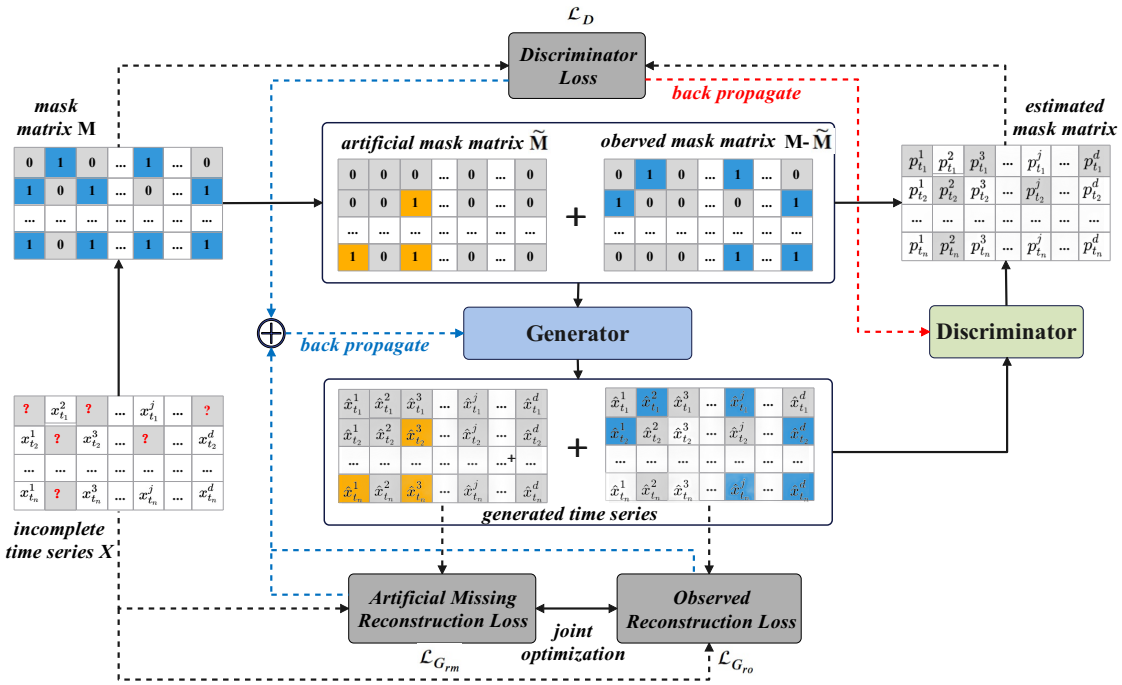


Fig. 4. The model training procedure.

let $\widetilde{\mathbf{M}}$ be the mask matrix indicating the artificially introduced missing values. If an observed value $x_{t_i}^j$ is artificially masked, then $\widetilde{m}_{t_i}^j = 1$; otherwise, $\widetilde{m}_{t_i}^j = 0$. The reconstruction loss is composed of two parts, as shown below.

$$\mathcal{L}_{G_{ro}} = \frac{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left[\left(m_{t_i}^j - \widetilde{m}_{t_i}^j \right) \cdot \left(x_{t_i}^j - \widehat{x}_{t_i}^j \right) \right]^2}{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left(m_{t_i}^j - \widetilde{m}_{t_i}^j \right)} \tag{17}$$

$$\mathcal{L}_{G_{rm}} = \frac{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left[\left(x_{t_i}^j - \widehat{x}_{t_i}^j \right) \cdot \widetilde{m}_{t_i}^j \right]^2}{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \widetilde{m}_{t_i}^j} \tag{18}$$

By introducing a certain number of synthetical missing values at observed locations, the model is compelled to focus more on these locations. This approach guides the model in learning to fill in the MVs, thereby enhancing its performance in MV imputation. Finally, the loss function of the generator G is defined below.

$$\mathcal{L}_G = \mathcal{L}_{G_d} + w_1 \cdot \mathcal{L}_{G_{ro}} + w_2 \cdot \mathcal{L}_{G_{rm}} \tag{19}$$

$$\mathcal{L}_{G_d} = -\frac{1}{nd} \sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} (1 - m_{t_i}^j) \cdot p_{t_i}^j \tag{20}$$

where w_1 and w_2 are the hyper-parameters used to control the weights of the reconstruction loss.(The experiment verifies that the model performance is stable when $w_1 = w_2$, and the effect is optimal when 100 is taken.) Additionally, it is important to note that with the joint optimization training strategy, the input to the generator becomes $\mathbf{X}_G \odot (\mathbf{M} - \widetilde{\mathbf{M}})$. In other words, some known information (observed values) becomes invisible to G, which may deteriorate the performance of MV imputation (as experimentally evaluated in Section V-B). Therefore, the artificial missing rate, which indicates how many observed values are manually masked, should not be set too high (we adopt 0.1 as the default value in this article).

Based on the loss functions introduced above, the generator G and the discriminator D are trained adversarially, ultimately achieving the desired imputation results. The joint-optimization model training procedure is shown in Figure 4 and the pseudo-code is presented in Algorithm 1.

E. Imputation

Given an incomplete time series $\mathbf{X}$, we try to reconstruct a complete time series $\widehat{\mathbf{X}}$ through the generator, ensuring that the generated $\widehat{\mathbf{X}}$ closely approximates the original time series. By using the joint-optimization training procedure, ultimately employ the corresponding values of $\widehat{\mathbf{X}}$ to estimate the missing values in $\mathbf{X}$. The final imputed time series is denoted as:

$$\widetilde{X} = \mathbf{X} \odot \mathbf{M} + \widehat{\mathbf{X}} \odot (1 - \mathbf{M}) \tag{21}$$

V. Experiments

A. Experimental Settings

Datasets: We adopt four real-world IoT time series datasets to conduct extensive experiments. The detailed information of each dataset is introduced below.

- **PhysioNet-2012 (PhysioNet 2012 Mortality Prediction Challenge)**: The dataset consists of approximately 12,000 ICU inpatient records, making it an IoT multivariate time series dataset. All patients in the ICU were admitted for reasons such as cardiac or internal medicine issues. Patient data collection begins 48 hours after admission to the ICU. A total of 37 time series variables are measured, including metrics like Urine Output, Glasgow Coma Score, and others. The measurements are both

Algorithm 1: AttnWGAIN

Input : incomplete time series $\mathbf{X}$, the mask matrix $\mathbf{M}$, artificial mask matrix $\widetilde{\mathbf{M}}$,hint matrix H and the time lag matrix Δ

Output: the imputed images $\widetilde{X}$

while *training loss has not converged* **do**
  foreach *batch with k_D data samples of training set* $\mathbf{X}$ **do**
    foreach $x^j_{t_i}$, $m^j_{t_i}$, $\widetilde{m^j_{t_i}}$,$H^j_{t_i}$, $\delta^j_{t_i}$, *in the current batch* **do**
      // Data preparation
      $\mathbf{z^j_{t_i}} \sim \mathcal{N}(0,1)$
      // Add noise
      $\mathbf{x}^j_{t_i} = \mathrm{m}^j_{t_i} \odot x^j_{t_i} + (1 - m^j_{t_i}) \odot \mathrm{z}^j_{t_i}$
      // Get generated time series and estimated mask matrix
      $\widehat{\mathbf{X}}^j_{t_i} = \text{Generator}\left(\mathbf{x}^j_{t_i}, m^j_{t_i}, \delta^j_{t_i}\right)$
      $\mathbf{P}^j_{t_i} = \text{Discriminator}\left(\mathbf{G}^j_{t_i}, H^j_{t_i}\right)$
      // calculate the loss of model
      $\mathcal{L}_D = \frac{1}{nd}\sum_{t_i=t_1}^{t_n}\sum_{j=1}^{d}\left[m^j_{t_i}\cdot p^j_{t_i} - (1-m^j_{t_i})\cdot p^j_{t_i}\right]$
      $\mathcal{L}_{G_{ro}} = \frac{\sum_{t_i=t_1}^{t_n}\sum_{j=1}^{d}\left[\left(m^j_{t_i}-\widetilde{m}^j_{t_i}\right)\cdot\left(x^j_{t_i}-\widehat{x}^j_{t_i}\right)\right]^2}{\sum_{t_i=t_1}^{t_n}\sum_{j=1}^{d}\left(m^j_{t_i}-\widetilde{m}^j_{t_i}\right)}$
      $\mathcal{L}_{G_{rm}} = \frac{\sum_{t_i=t_1}^{t_n}\sum_{j=1}^{d}\left[\left(x^j_{t_i}-\widehat{x}^j_{t_i}\right)\cdot\widetilde{m}^j_{t_i}\right]}{\sum_{t_i=t_1}^{t_n}\sum_{j=1}^{d}\widetilde{m}^j_{t_i}}$
      $\mathcal{L}_{G_d} = -\frac{1}{nd}\sum_{t_i=t_1}^{t_n}\sum_{j=1}^{d}(1-m^j_{t_i})\cdot p^j_{t_i}$
      $\mathcal{L}_G = \mathcal{L}_{G_d} + w_1\cdot\mathcal{L}_{G_{ro}} + w_2\cdot\mathcal{L}_{G_{rm}}$
    // update the model
    $\bigtriangledown_D \sum_{i=1}^{k_D}\mathcal{L}_D\left(\mathbf{M}^j_{t_i}, \mathbf{P}^j_{t_i}\right)\mathcal{L}_G$
    $\bigtriangledown_G \sum_{i=1}^{k_D}\mathcal{L}_G\left(\mathbf{X}^j_{t_i}, \overline{\mathbf{X}^j_{t_i}}, \mathbf{M}^j_{t_i}, \widetilde{\mathbf{M}}^j_{t_i}\right)$
// missing value imputation
$\widetilde{X} = \mathbf{X}\odot\mathbf{M} + \widehat{\mathbf{X}}\odot(1-\mathbf{M})$
Return $\widetilde{X}$

regular and irregular, and not all variables are available for every patient sample. It is important to note that this is a high missing rate dataset, with approximately 80% of the data missing. In our study, we split the dataset into training and testing sets in an 80:20 ratio. In addition, we allocated 20% of the training data for validation purposes. To assess the imputation performance of our model, we intentionally masked 10% of the observations in both the validation and test sets, treating these masked entries as the true values for evaluation.

- **Air-Quality (Beijing Multi-Site Air-Quality)**: This dataset contains hourly measurements of air pollutants from 12 nationally-monitored air-quality stations. The data on air quality is sourced from the Beijing Municipal Environmental Monitoring Center. Meteorological data for each air-quality station is paired with the nearest weather station provided by the China Meteorological Administration. The time period is from March 1st, 2013 to February 28th, 2017. Each monitoring station measured 11 continuous time series variables (such as NO2, CO, O3, etc.). However, the measurement times for each pollutant vary; some are daily averages, while others are hourly averages. We aggregated the variables from the 12 stations, resulting in a dataset with 132 features and 1.6% missing values. The dataset covers a total of 48 months of data. We used the first 10 months (from March 2013 to December 2013) as the test set, the next 10 months (from January 2014 to October 2014) as the validation set, and the remaining 28 months as the training set. Additionally, we treated each consecutive 24 time steps as a sample to generate time series data samples. Similar to the PhysioNet-2012 dataset, 10% of the data in both the validation and test sets will be artificially masked to assess the model's imputation performance.
- **Electricity (Electricity Load Diagrams)**: This is an open-source public dataset from UCI [62]. It includes electricity consumption data (in kilowatt-hours) collected every 15 minutes from 370 customers. The dataset is complete, with no missing values. The data spans from January 1, 2011, to December 31, 2014, covering a total of 48 months. We designated the first 10 months (from January 2011 to October 2011) as the test set, the next 10 months (from November 2011 to August 2012) as the validation set, and the remaining 28 months as the training set. Additionally, we treat every consecutive 100 time steps as a sample to generate time series data samples. Since the dataset has no missing values, we mask 10% of the observations in both the validation and test sets and use them to evaluate the model's imputation performance.
- **ETT (Electricity Transformer Temperature)**: The ETT dataset [63] is the online monitoring data of 110 kV power transformers collected by Beihang University and Beijing State Grid Fuda Technology Development Company over a period of two years. We selected a portion of this dataset, namely ETTm1, as our research dataset. The ETTm1 dataset covers seven status variables such as load and temperature rise, which reflect the operating conditions of the transformer and provide researchers with a multi-angle observation and analysis perspective. Similar to the processing of Electricity, we masked 10% of the observations in the validation set and test set and used them as the basis for evaluation of the ground truth.

Algorithms for comparison: We compare the proposed AttnWGAIN with nine representative existing time series imputation algorithms, i.e., Median, Last, GRUI-GAN [28], E^2GAN [29], M-RNN [38], GP-VAE [42], BRITS [34], Transformer [51], SAITS [25]. The Meaian and Last are two native imputation methods where the MVs are replaced by the median values of the training set and the last observed values, respectively. GRUI-GAN and E^2GAN are state-of-the-art GAN-based time series imputation methods proposed by Yoon et al. The M-RNN, GP-VAE, and BRITS [34] are autoregressive imputation models that learn temporal correlations based on the RNN model. The Transformer and SAITS are the latest attention-based time series imputation models, where the input data is processed parallelly and globally.

Model Settings: We implement the proposed AttnWGAIN using PyTorch, drawing extensively on prior knowledge for the model's construction. While the model performs well

TABLE II
GENERAL INFORMATION ABOUT THE FOUR DATASETS USED IN THIS WORK

	PhysioNet-2012	Air-Quality	Electricity	ETT(m1)
Number of total samples	11,988	1,461	1,400	69680
Number of features	37	132	370	7
Sequence length	48	24	100	24
Original missing rate	80.67%	1.6%	0%	0%

TABLE III
HYPER-PARAMETER1 SETTINGS

	PhysioNet-2012	Air-Quality	Electricity	ETT(m1)
$alpha$	[100, 100]	[100, 100]	[100, 100]	[100, 100]
$batch_size$	128	128	128	128
d_inner	512	512	128	512
d_k	32	128	128	32
d_model	256	2048	2048	256
d_v	32	128	128	32
$dropout$	0	0	0	0
$epochs$	1000	10000	2000	10000
lr	0.0005	0.0001	0.0002	0.001
$artificial_miss_rate$	0.1	0.1	0.1	0.1
$hint_rate$	0.1	0.1	0.1	0.1
$n_group_inner_layers$	1	1	1	5
n_groups	5	1	1	1
n_head	8	4	8	1

with the final parameters set as follows, there remains room for optimization. The hyperparameter settings of AttnWGAIN are detailed in Table III. Here, $alpha$ corresponds to w_1 and w_2 in the text. The parameter n_groups denotes the number of stacked encoding blocks in the generator, $n_group_inner_layers$ controls the number of repeated applications within each encoder group, and n_head represents the number of heads in the multi-head attention mechanism. The experimental results for the comparison methods on the first three datasets are sourced from SAITS, whereas the results for the ETT dataset are obtained by running the source code provided by the original authors. All models are trained with the Adam optimizer on NVIDIA RTX A6000 GPUs.

Measurements: We adopt three measurements, i.e., MAE (Mean Absolute Error), RMSE (Root Mean Square Error), and MRE (Mean Relative Error), to evaluate the MV imputation performance across various models. Note that imputation errors are computed only for the missing values indicated by the mask matrix $\mathbf{M}$. The mathematical formulation for each metric is provided below.

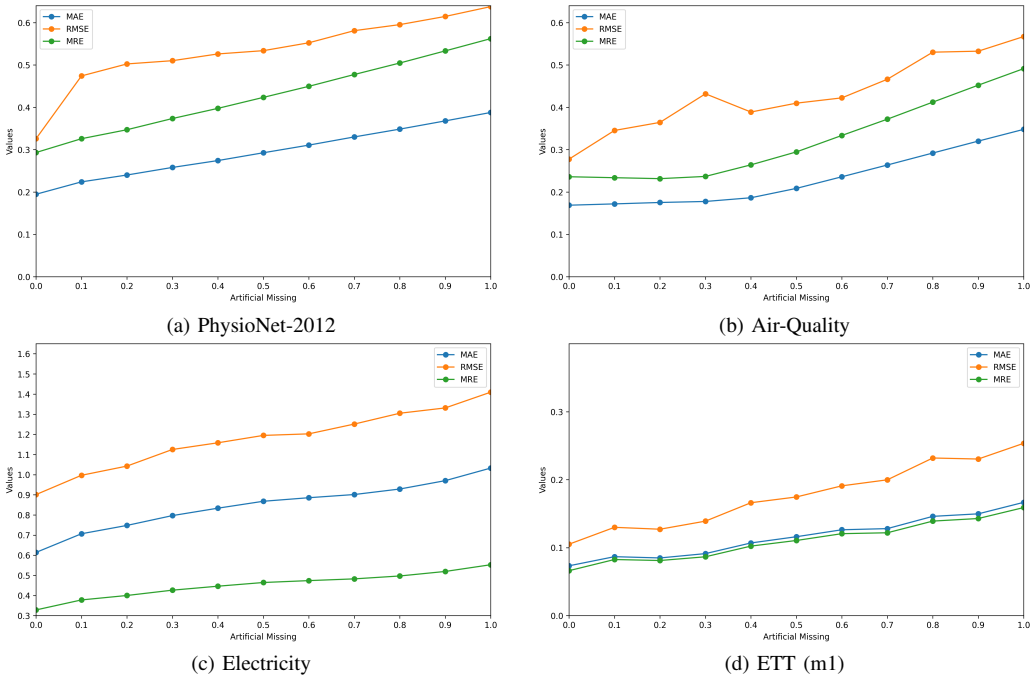
$$\text{MAE}(\mathbf{X}, \widetilde{\mathbf{X}}, \mathbf{M}) = \frac{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left| \left(x_{t_i}^j - \widetilde{x}_{t_i}^j \right) \cdot m_{t_i}^j \right|}{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} m_{t_i}^j} \tag{22}$$

$$\text{RMSE}(\mathbf{X}, \widetilde{\mathbf{X}}, \mathbf{M}) = \sqrt{\frac{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left[\left(x_{t_i}^j - \widetilde{x}_{t_i}^j \right)^2 \cdot m_{t_i}^j \right]}{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} m_{t_i}^j}} \tag{23}$$

$$\text{MRE}(\mathbf{X}, \widetilde{\mathbf{X}}, \mathbf{M}) = \frac{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left| \left(x_{t_i}^j - \widetilde{x}_{t_i}^j \right) \cdot m_{t_i}^j \right|}{\sum_{t_i=t_1}^{t_n} \sum_{j=1}^{d} \left| \widetilde{x}_{t_i}^j \cdot m_{t_i}^j \right|} \tag{24}$$

B. Selection of Artificial Missing Rate

As introduced in Section IV-D, we manually mask a certain part of observed values to make the generator G to focus more on reconstructing missing values. However, this joint-optimization training strategy can be a double-edged sword, potentially degrading the final imputation performance because some available information becomes inaccessible. As illustrated in Fig. 5, without introducing artificial missing values (i.e., an artificial missing rate of 0%), the imputation performance is noticeably worse compared to scenarios with artificial missing values. Conversely, as the artificial missing rate increases, the performance of AttnWGAIN declines. Therefore, selecting an appropriate artificial missing rate is crucial to balancing imputation performance. In this study, we set the default artificial missing rate to 10%.



(a) PhysioNet-2012 (b) Air-Quality

(c) Electricity (d) ETT (m1)

Fig. 5. Performance of different artificial missing rates on four datasets

C. Comparison with Baselines

First, we present the imputation errors using three measurements (MAE/RMSE/MRE) across four datasets with a missing rate of 10%. As shown in Table IV, recent SOTA deep-learning-based imputation models generally outperform the Median and Last methods, confirming the effectiveness of deep learning models for time series imputation. GRUI-GAN and E2GAN, which are implemented with traditional GANs, struggle with loss explosion when applied to the Electricity dataset with a long sequence length of 100. In contrast, the proposed AttnWGAIN, implemented with WGAN, performs favorably and achieves satisfactory results. Additionally, compared to solely GAN-based models (i.e., GRUI-GAN and E^2GAN) and RNN-based models (i.e., M-RNN, GP-VAE, and BRITS), attention-based models (i.e., Transformer and SAITS) generally perform better in most cases. This advantage may be attributed to the ability of attention-based models to dynamically focus and filter information, making them more adept at handling long-sequence time series. Finally, by integrating the attention mechanism with GANs, AttnWGAIN significantly outperforms all baseline models across all datasets.

Next, to further demonstrate the superiority of AttnWGAIN, Table V presents the imputation performance of various models across different missing rates (20% $\sim$ 90%) on the Electricity dataset. We have omitted GRUI-GAN and E^2GAN from Table V, as their imputation results could not be obtained due to loss explosion. It is unsurprising that the imputation errors of all deep-learning imputation models increase gradually with higher missing rates, as less available information can be used for MV imputation. Consistent with the results shown in Table V, attention-based imputation models outperform others, and the proposed AttnWGAIN achieves the best performance in eight out of nine cases, reaffirming its advantage.

D. Applications in Classification

Due to the pervasive existence of MVs in real-world datasets, tasks such as classification and pattern recognition become challenging. To evaluate the benefits AttnWGAIN can offer for classification, we applied it to a real-world data classification task using the PhysioNet-2012 dataset. The core idea of the experimental design is that if the overall quality of the input data is superior, this high-quality dataset should enhance performance in subsequent classification tasks. We implemented an RNN-based classifier with settings consistent with those in SAITS [25]. Table VI reports the classification results using three metrics: ROC-AUC [64], PR-AUC [65], and F1-score [66]), for data without imputation (denoted as "missing" in the table), data filled by existing methods, and data processed by AttnWGAIN. The results show that classification accuracy is lowest without imputation, largely because the missing information is crucial to classification outcomes. Since classification accuracy generally correlates with imputation quality, AttnWGAIN consistently achieves the best performance across all metrics, with explicit improvements of 0.5%, 3.7%, and 1.3% in ROC-AUC, PR-AUC, and F1-score, respectively, compared to the latest state-of-the-art SAITS model. These results demonstrate the superiority and general applicability of AttnWGAIN.

E. Ablation Study

To enhance imputation accuracy, AttnWGAIN incorporates several meticulous designs: (1) including time lag information Δ as part of the input; (2) utilizing a joint optimization training strategy, which involves manually masking some observed values during generator training; and (3) employing diagonally masked multi-head attention for generative imputation. In this subsection, we present ablation experiments to demonstrate

TABLE IV
PERFORMANCE COMPARISON FOR IMPUTATION

Method	PhysioNet-2012	Air-Quality	Electricity	ETT(m1)
Median	0.725 / 0.985 / 103.1%	0.761 / 1.173 / 107.2%	2.088 / 2.739 / 112.1%	1.151 / 1.862 / 139.9%
Last	0.850 / 1.188 / 118.3%	0.948 / 1.389 / 129.1%	0.988 / 1.497 / 51.3%	0.977 / 1.257 / 95.9%
GRUI-GAN	0.760 / 1.021 / 107.0%	0.766 / 1.155 / 108.1%	/	0.603 / 0.710 / 94.2%
E^2GAN	0.698 / 0.944 / 99.1%	0.752 / 1.124 / 105.7%	/	0.558 / 0.688 / 87.0%
M-RNN	0.521 / 0.765 / 74.9%	0.302 / 0.651 / 42.0%	1.231 / 1.848 / 65.8%	0.389 / 0.446 / 32.1%
GP-VAE	0.410 / 0.655 / 57.1%	0.273 / 0.632 / 38.3%	1.113 / 1.591 / 58.9%	0.277 / 0.322 / 15.9%
BRITS	0.259 / 0.780 / 36.5%	0.159 / 0.554 / 22.6%	0.881 / 1.333 / 46.3%	0.133 / 0.264 / 12.8%
Transformer	0.197 / 0.466 / 27.5%	0.168 / 0.571 / 23.3%	0.877 / 1.327 / 45.6%	0.121 / 0.193 / 11.8%
SAITS	0.186 / 0.431 / **26.6%**	0.143 / 0.336 / 20.3%	0.796 / 1.232 / 42.6%	0.098 / 0.151 / 9.3%
AttnWGAIN	**0.179 / 0.314** / 26.9%	**0.138 / 0.330 / 19.6%**	**0.593 / 0.826 / 31.8%**	**0.078 / 0.116 / 7.5%**

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE ELECTRICITY DATASET ACROSS VARYING MISSING RATES FROM 20% TO 90% IS PRESENTED. THE REPORTED METRICS ARE MAE / RMSE / MRE, WHERE LOWER VALUES INDICATE BETTER PERFORMANCE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

Method	20%	30%	40%	50%
Median	2.056 / 2.732 / 110.1%	2.059 / 2.734 / 110.2%	2.060 / 2.736 / 110.3%	2.052 / 2.727 / 109.8%
Last	1.013 / 1.548 / 54.3%	1.020 / 1.562 / 54.6%	1.023 / 1.575 / 54.7%	1.030 / 1.594 / 55.1%
M-RNN	1.239 / 1.851 / 66.3%	1.260 / 1.877 / 67.4%	1.271 / 1.883 / 68.1%	1.291 / 1.905 / 68.9%
GP-VAE	1.120 / 1.497 / 59.9%	1.012 / 1.540 / 56.2%	1.090 / 1.567 / 58.1%	1.089 / 1.567 / 58.6%
BRITS	0.933 / 1.412 / 49.9%	0.942 / 1.433 / 50.4%	0.988 / 1.489 / 53.9%	1.021 / 1.528 / 55.1%
Transformer	0.851 / 1.325 / 45.4%	0.857 / 1.331 / 45.6%	0.878 / 1.389 / 46.9%	0.901 / 1.418 / 48.2%
SAITS	0.763 / 1.187 / 40.8%	0.790 / 1.223 / 42.3%	0.869 / 1.314 / 46.7%	0.876 / 1.377 / 46.9%
AttnWGAIN	**0.596 / 0.822 / 31.9%**	**0.645 / 0.882 / 34.6%**	**0.630 / 0.868 / 33.7%**	**0.701 / 0.961 / 37.5%**
Method	60%	70%	80%	90%
Median	2.055 / 2.732 / 110.1%	2.054 / 2.730 / 109.9%	2.060 / 2.735 / 110.3%	2.055 / 2.724 / 110.2%
Last	1.042 / 1.617 / 55.8%	1.046 / 1.642 / 56.1%	1.058 / 1.661 / 56.6%	1.071 / 1.692 / 57.3%
M-RNN	1.295 / 1.910 / 69.3%	1.307 / 1.931 / 70.0%	1.320 / 1.961 / 70.6%	1.337 / 1.975 / 72.1%
GP-VAE	1.103 / 1.621 / 59.1%	1.042 / 1.601 / 55.7%	1.068 / 1.629 / 57.0%	1.008 / 1.623 / 53.9%
BRITS	1.103 / 1.606 / 59.1%	1.092 / 1.611 / 58.6%	1.143 / 1.667 / 61.2%	1.170 / 1.713 / 62.8%
Transformer	0.898 / 1.412 / 48.1%	0.908 / 1.432 / 48.7%	0.921 / 1.468 / 49.4%	0.942 / 1.503 / 50.2%
SAITS	0.892 / 1.328 / 47.9%	0.898 / 1.273 / 48.1%	0.908 / 1.327 / 48.6%	0.933 / 1.354 / 49.9%
AttnWGAIN	**0.686 / 0.930 / 36.8%**	**0.757 / 0.991 / 40.6%**	**0.871 / 1.113 / 46.6%**	**0.817 / 1.109 / 43.7%**

TABLE VI
THE RESULTS OF THE DOWNSTREAM CLASSIFICATION TASK ON THE PHYSIONET-2012 DATASET ARE PRESENTED. PERFORMANCE METRICS FOR EACH METHOD ARE BASED ON FIVE INDEPENDENT RUNS, WITH THE REPORTED VALUES REPRESENTING THE MEANS ± STANDARD DEVIATIONS. HIGHER VALUES INDICATE BETTER PERFORMANCE.

Method	ROC-AUC	PR-AUC	F1-score
Median	83.3% ± 0.4%	46.2% ± 0.5%	38.3% ± 2.9%
Last	82.6% ± 0.3%	46.7% ± 0.5%	39.3% ± 2.2%
GRUI-GAN	82.9% ± 0.3%	45.1% ± 0.8%	38.5% ± 2.1%
E^2GAN	83.0% ± 0.2%	45.0% ± 0.7%	36.1% ± 2.2%
M-RNN	82.4% ± 0.2%	45.3% ± 0.5%	38.5% ± 3.2%
GP-VAE	83.5% ± 0.2%	47.9% ± 0.8%	40.6% ± 3.4%
BRITS	83.6% ± 0.1%	49.0% ± 0.5%	41.6% ± 1.6%
Transformer	84.2% ± 0.7%	49.1% ± 1.5%	41.3% ± 2.0%
SAITS	84.6% ± 0.4%	50.9% ± 0.7%	42.6% ± 2.8%
AttnWGAIN	**85.3% ± 0.3%**	**54.7% ± 0.6%**	**44.0% ± 1.2%**

TABLE VII
ABLATION EXPERIMENT RESULTS WITH EACH PART OF MODIFICATIONS

Model	PhysioNet-2012	Air-Quality	Electricity	ETT(m1)
AttnWGAIN w/o (Δ, DiagMask, JOT)	0.578 / 0.7573 / 87.1%	0.264 / 0.463 / 37.3%	0.929 / 1.465 / 49.7%	0.617 / 0.888 / 58.9%
AttnWGAIN w/o (Δ, JOT)	0.580 / 0.759 / 87.4%	0.261 / 0.464 / 36.8%	0.928 / 1.465 / 49.6%	0.620 / 0.889 / 59.1%
AttnWGAIN w/o JOT	0.546 / 0.723 / 82.2%	0.258 / 0.457 / 36.4%	0.942 / 1.463 / 50.4%	0.651 / 0.937 / 62.0%
AttnWGAIN w/o Δ	0.183 / 0.318 / 27.5%	0.140 / **0.324** / 19.8%	0.613 / 0.830 / 32.7%	0.088 / 0.133 / 8.4%
AttnWGAIN	**0.179 / 0.314 / 26.9%**	**0.138** / 0.330 / **19.6%**	**0.593 / 0.826 / 31.8%**	**0.078 / 0.116 / 7.5%**

TABLE VIII
ABLATION EXPERIMENT RESULTS WITH DIFFERENT NETWORK ARCHITECTURE OF D

Model	PhysioNet-2012	Air-Quality	Electricity	ETT(m1)
AttnWGAIN	0.179 / 0.314 / 26.9%	0.138 / 0.330 / 19.6%	0.593 / 0.826 / 31.8%	0.078 / 0.116 / 7.5%
AttnWGAIN-AttnD	0.178 / 0.310 / 26.8%	0.151 / 0.341 / 21.5%	0.616 / 0.860 / 33.0%	0.080 / 0.119 / 7.6%

the effectiveness of each design in improving missing value (MV) imputation accuracy. Table VII presents the experimental results on four real-world datasets. The results show that the baseline model, which does not incorporate any of the designs (AttnWGAIN w/o (Δ, DiagMask, JOT) in Table VII), exhibits the poorest performance. By incrementally adding each module to the base model, we observe that the insertion of modules—whether JOT, DiagMask, or Δ—significantly improves imputation accuracy. This progression culminates in AttnWGAIN achieving optimal performance, confirming that each design contributes to enhancing imputation accuracy.

Additionally, we implemented another version of AttnWGAIN (denoted as AttnWGAIN-AttnD), where the network architecture of the discriminator D is the same as that of the generator G. As shown in Table VIII, the performance of AttnWGAIN with a simple architecture (i.e., two fully connected layers) is comparable to that of AttnWGAIN-AttnD, while AttnWGAIN remains more efficient with fewer parameters.

VI. CONCLUSION

In this paper, we propose AttnWGAIN, a GAN-based model that improves the generator in WGAN by leveraging a series of diagonally masked attention blocks to reconstruct complete time-series data. The discriminator aims to distinguish between real (observed) and fake (imputed) elements to enhance imputation accuracy. Additionally, we introduce a time lag matrix into the generator's input to explicitly model dynamic temporal correlations and a hint matrix into the discriminator's input to assist in distinguishing real and fake elements. Training the model with a joint optimization strategy enhances AttnWGAIN's competitiveness in time-series imputation.

Compared with SAITS, a state-of-the-art self-attention-based imputation model, AttnWGAIN demonstrates significant improvements across four real-world datasets: mean absolute error (MAE) is reduced by 5% to 26%, root mean square error (RMSE) by 3% to 33%, and mean relative error (MRE) by more than 15% on average. AttnWGAIN effectively achieves state-of-the-art accuracy in time-series imputation tasks. Furthermore, empirical results from downstream classification experiments provide qualitative evidence that imputing missing data with AttnWGAIN directly improves classification performance.

Future research will focus on exploring the imputation performance of the AttnWGAIN model in time series data with different missing patterns, while introducing more diversified evaluation metrics to achieve a comprehensive quantitative analysis of imputation performance. Furthermore, future studies will incorporate more mainstream generative model architectures (such as diffusion models) for comparison and integration, further enhancing the model's performance in time series imputation tasks.

ACKNOWLEDGMENTS

This work is partly supported by the Fundamental Research Funds for the Central Universities (Grant No. 3132024269, 3132024266, 3132023523), the National Natural Science Foundation of China (Grant Nos. 62002039).

REFERENCES

[1] L. Zhenxin, Z. Fengming, W. Ying, T. Qian, and L. Chao, "Method of missing data imputation for multivariate time series," *Systems Engineering and Electronics*, vol. 40, no. 01, pp. 225–230, 2018.

[2] B. Singh, P. Kumar, N. Sharma, and K. Sharma, "Sales forecast for amazon sales with time series modeling," in *2020 first international conference on power, control and computing technologies (ICPC2T)*. IEEE, 2020, pp. 38–43.

[3] A. Shrieenidhi, P. Ranjana *et al.*, "Walmart sales forecasting using time series analysis," in *2024 International Conference on Advances in Data Engineering and Intelligent Computing Systems (ADICS)*. IEEE, 2024, pp. 1–5.

[4] Q. Lin, H. Kuang, and X. Zhang, "Design and implementation of big data forecasting system based on intelligent transportation," *IEEE Trans. Consumer Electron.*, vol. 70, no. 1, pp. 1286–1293, 2024. [Online]. Available: https://doi.org/10.1109/TCE.2023.3319639

[5] P. Chatterjee, D. Das, and D. B. Rawat, "Federated learning empowered recommendation model for financial consumer services," *IEEE Trans. Consumer Electron.*, vol. 70, no. 1, pp. 2508–2516, 2024. [Online]. Available: https://doi.org/10.1109/TCE.2023.3339702

[6] Y. Lai and M. Chen, "Using natural language processing with explainable AI approach to construct a human-centric consumer application for financial climate disclosures," *IEEE Trans. Consumer Electron.*, vol. 70, no. 1, pp. 1112–1121, 2024. [Online]. Available: https://doi.org/10.1109/TCE.2023.3326953

[7] E. Choi, C. Xiao, W. Stewart, and J. Sun, "Mime: Multilevel medical embedding of electronic health records for predictive healthcare," *Advances in neural information processing systems*, vol. 31, 2018.

[8] M. Horn, M. Moor, C. Bock, B. Rieck, and K. Borgwardt, "Set functions for time series," in *International Conference on Machine Learning*. PMLR, 2020, pp. 4353–4363.

[9] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-c. Woo, "Convolutional lstm network: A machine learning approach for precipitation nowcasting," *Advances in neural information processing systems*, vol. 28, 2015.

[10] X. Yi, Y. Zheng, J. Zhang, and T. Li, "St-mvl: Filling missing values in geo-sensory time series data," in *Proceedings of the 25th International Joint Conference on Artificial Intelligence*, 2016.

[11] L. Ying and X. Yunlong, "Research on public opinion monitoring of blockchain industry that integrates sentiment analysis and multiple time series," *Technology Intelligence Engineering*, vol. 9, no. 01, pp. 3–14, 2023.

[12] H. Nugroho and K. Surendro, "Missing data problem in predictive analytics," in *Proceedings of the 2019 8th International Conference on Software and Computer Applications*, 2019, pp. 95–100.

[13] W. Gao and Z. Zhao, "Self-supervised multi-source heterogeneous data fusion using encode and decode attention for intelligent medical device communication analysis," *IEEE Trans. Consumer Electron.*, vol. 70, no. 1, pp. 1318–1325, 2024. [Online]. Available: https://doi.org/10.1109/TCE.2023.3321331

[14] R. Baker *et al.*, "Data mining for education," *International encyclopedia of education*, vol. 7, no. 3, pp. 112–118, 2010.

[15] Z.-g. Liu, Q. Pan, J. Dezert, and A. Martin, "Adaptive imputation of missing values for incomplete pattern classification," *Pattern Recognition*, vol. 52, pp. 85–95, 2016.

[16] S. M. Fox-Wasylyshyn and M. M. El-Masri, "Handling missing data in self-report measures," *Research in nursing & health*, vol. 28, no. 6, pp. 488–495, 2005.

[17] D. J. Stekhoven and P. Bühlmann, "Missforest—non-parametric missing value imputation for mixed-type data," *Bioinformatics*, vol. 28, no. 1, pp. 112–118, 2012.

[18] Z. Songlan, W. Peng, and X. Ziwei, "A study of missing value data processing based on statistical correlation," *Statistics Decision*, pp. 13–16, 2016.

[19] M. Templ, A. Kowarik, and P. Filzmoser, "Iterative stepwise regression imputation using standard and robust methods," *Computational Statistics & Data Analysis*, vol. 55, no. 10, pp. 2793–2806, 2011.

[20] R. Malarvizhi and A. S. Thanamani, "K-nearest neighbor in missing data imputation," *Int. J. Eng. Res. Dev*, vol. 5, no. 1, pp. 5–7, 2012.

[21] F. Tang and H. Ishwaran, "Random forest missing data algorithms," *Statistical Analysis and Data Mining: The ASA Data Science Journal*, vol. 10, no. 6, pp. 363–377, 2017.

[22] T. Sivapriya, A. Kamal, and V. Thavavel, "Imputation and classification of missing data using least square support vector machines–a new approach in dementia diagnosis," *Int. J. Adv. Res. Artif. Intell*, vol. 1, no. 4, pp. 29–33, 2012.

[23] Z. Chan, "A support vector machine-based missing values filling algorithm," *Computer Applications and Software*, vol. 30, no. 05, pp. 226–228, 2013.

[24] X. Wang, X. Liu, W. Pedrycz, and L. Zhang, "Fuzzy rule based decision trees," *Pattern Recognition*, vol. 48, no. 1, pp. 50–59, 2015.

[25] W. Du, D. Côté, and Y. Liu, "Saits: Self-attention-based imputation for time series," *Expert Systems with Applications*, vol. 219, p. 119619, 2023.

[26] J. Yoon, J. Jordon, and M. Schaar, "Gain: Missing data imputation using generative adversarial nets," in *International conference on machine learning*. PMLR, 2018, pp. 5689–5698.

[27] Z. Wu, X. Jia, R. Jiang, Y. Ye, H. Qi, and C. Xu, "CSID-GAN: A customized style interior floor plan design framework based on generative adversarial network," *IEEE Trans. Consumer Electron.*, vol. 70, no. 1, pp. 2353–2364, 2024. [Online]. Available: https://doi.org/10.1109/TCE.2024.3376956

[28] Y. Luo, X. Cai, Y. Zhang, J. Xu *et al.*, "Multivariate time series imputation with generative adversarial networks," *Advances in neural information processing systems*, vol. 31, 2018.

[29] Y. Luo, Y. Zhang, X. Cai, and X. Yuan, "E2gan: End-to-end generative adversarial network for multivariate time series imputation," in *Proceedings of the 28th international joint conference on artificial intelligence*. AAAI Press Palo Alto, CA, USA, 2019, pp. 3094–3100.

[30] S. Yoon and S. Sull, "Gamin: Generative adversarial multiple imputation network for highly missing data," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 8456–8464.

[31] L. Xu, Z. Song, D. Wang, J. Su, Z. Fang, C. Ding, W. Gan, Y. Yan, X. Jin, X. Yang *et al.*, "Actformer: A gan-based transformer towards general action-conditioned 3d human motion generation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 2228–2238.

[32] Y.-S. Wang, C.-Y. Chen, and L.-H. Lee, "Generative adversarial networks based on mixed-attentions for citation intent classification in scientific publications," in *Proceedings of the 33rd Conference on Computational Linguistics and Speech Processing (ROCLING 2021)*, 2021, pp. 280–285.

[33] Y. Liu, R. Yu, S. Zheng, E. Zhan, and Y. Yue, "Naomi: Non-autoregressive multiresolution sequence imputation," *Advances in neural information processing systems*, vol. 32, 2019.

[34] W. Cao, D. Wang, J. Li, H. Zhou, L. Li, and Y. Li, "Brits: Bidirectional recurrent imputation for time series," *Advances in neural information processing systems*, vol. 31, 2018.

[35] Y. Zhang, Y. Liu, Y. Xu, H. Xiong, C. Lei, W. He, L. Cui, and C. Miao, "Enhancing sequential recommendation with graph contrastive learning," 2022. [Online]. Available: https://arxiv.org/abs/2205.14837

[36] Y. Yang, C. Huang, L. Xia, C. Huang, D. Luo, and K. Lin, "Debiased contrastive learning for sequential recommendation," in *Proceedings of the ACM Web Conference 2023*, ser. WWW '23. ACM, Apr. 2023, p. 1063–1073. [Online]. Available: http://dx.doi.org/10.1145/3543507.3583361

[37] Z. Che, S. Purushotham, K. Cho, D. Sontag, and Y. Liu, "Recurrent neural networks for multivariate time series with missing values," *Scientific reports*, vol. 8, no. 1, p. 6085, 2018.

[38] J. Yoon, W. R. Zame, and M. van der Schaar, "Estimating missing data in temporal data streams using multi-directional recurrent neural networks," *IEEE Transactions on Biomedical Engineering*, vol. 66, no. 5, pp. 1477–1490, 2018.

[39] A. Cini, I. Marisca, and C. Alippi, "Multivariate time series imputation by graph neural networks. corr abs/2108.00298 (2021)," *arXiv preprint arXiv:2108.00298*, 2021.

[40] I. Marisca, A. Cini, and C. Alippi, "Learning to reconstruct missing data from spatiotemporal graphs with sparse observations," *Advances in Neural Information Processing Systems*, vol. 35, pp. 32 069–32 082, 2022.

[41] P.-A. Mattei and J. Frellsen, "Miwae: Deep generative modelling and imputation of incomplete data sets," in *International conference on machine learning*. PMLR, 2019, pp. 4413–4423.

[42] V. Fortuin, D. Baranchuk, G. Rätsch, and S. Mandt, "Gp-vae: Deep probabilistic time series imputation," in *International conference on artificial intelligence and statistics*. PMLR, 2020, pp. 1651–1661.

[43] A. Nazabal, P. M. Olmos, Z. Ghahramani, and I. Valera, "Handling incomplete heterogeneous data using vaes," *Pattern Recognition*, vol. 107, p. 107501, 2020.

[44] S. Ramchandran, G. Tikhonov, K. Kujanpää, M. Koskinen, and H. Lähdesmäki, "Longitudinal variational autoencoder," in *International Conference on Artificial Intelligence and Statistics*. PMLR, 2021, pp. 3898–3906.

[45] M. Ashman, J. So, W. Tebbutt, V. Fortuin, M. Pearce, and R. E. Turner, "Sparse gaussian process variational autoencoders," *arXiv preprint arXiv:2010.10177*, 2020.

[46] Q. Ma, X. Li, M. Bai, X. Wang, B. Ning, and G. Li, "Mivae: multiple imputation based on variational auto-encoder," *Engineering Applications of Artificial Intelligence*, vol. 123, p. 106270, 2023.

[47] Y. Tashiro, J. Song, Y. Song, and S. Ermon, "Csdi: Conditional score-based diffusion models for probabilistic time series imputation," *Advances in Neural Information Processing Systems*, vol. 34, pp. 24 804–24 816, 2021.

[48] S. Zheng and N. Charoenphakdee, "Diffusion models for missing value imputation in tabular data," *arXiv preprint arXiv:2210.17128*, 2022.

[49] M. Liu, H. Huang, H. Feng, L. Sun, B. Du, and Y. Fu, "Pristi: A conditional diffusion framework for spatiotemporal imputation," in *2023 IEEE 39th International Conference on Data Engineering (ICDE)*. IEEE, 2023, pp. 1927–1939.

[50] J. M. L. Alcaraz and N. Strodthoff, "Diffusion-based time series imputation and forecasting with structured state space models," *arXiv preprint arXiv:2208.09399*, 2022.

[51] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[52] J. Ma, Z. Shou, A. Zareian, H. Mansour, A. Vetro, and S.-F. Chang, "Cdsa: cross-dimensional self-attention for multivariate, geo-tagged time series imputation," *arXiv preprint arXiv:1905.09904*, 2019.

[53] P. Bansal, P. Deshpande, and S. Sarawagi, "Missing value imputation on multidimensional time series," *arXiv preprint arXiv:2103.01600*, 2021.

[54] A. Venkatraman, M. Hebert, and J. Bagnell, "Improving multi-step prediction of learned time series models," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 29, no. 1, 2015.

[55] R. Deng, B. Chang, M. A. Brubaker, G. Mori, and A. Lehrmann, "Modeling continuous stochastic processes with dynamic normalizing flows," *Advances in Neural Information Processing Systems*, vol. 33, pp. 7805–7815, 2020.

[56] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training gans," *Advances in neural information processing systems*, vol. 29, 2016.

[57] A. Lugmayr, M. Danelljan, A. Romero, F. Yu, R. Timofte, and L. Van Gool, "Repaint: Inpainting using denoising diffusion probabilistic models," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11 461–11 471.

[58] R. Xiong, Y. Yang, D. He, K. Zheng, S. Zheng, C. Xing, H. Zhang, Y. Lan, L. Wang, and T. Liu, "On layer normalization in the transformer architecture," in *International Conference on Machine Learning*. PMLR, 2020, pp. 10 524–10 533.

[59] Q. Wang, B. Li, T. Xiao, J. Zhu, C. Li, D. F. Wong, and L. S. Chao, "Learning deep transformer models for machine translation," *arXiv preprint arXiv:1906.01787*, 2019.

[60] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial networks," *Communications of the ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[61] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *International conference on machine learning*. PMLR, 2017, pp. 214–223.

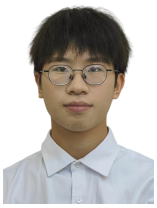
[62] A. Trindade, "ElectricityLoadDiagrams20112014," UCI Machine Learning Repository, 2015, DOI: https://doi.org/10.24432/C58C86.

[63] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *The Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021, Virtual Conference*, vol. 35, no. 12. AAAI Press, 2021, pp. 11 106–11 115.

[64] A. M. Carrington, D. G. Manuel, P. W. Fieguth, T. Ramsay, V. Osmani, B. Wernly, C. Bennett, S. Hawken, O. Magwood, Y. Sheikh *et al.*, "Deep roc analysis and auc as balanced average accuracy, for improved classifier selection, audit and explanation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 1, pp. 329–341, 2022.

[65] H. R. Sofaer, J. A. Hoeting, and C. S. Jarnevich, "The area under the precision-recall curve as a performance metric for rare binary events," *Methods in Ecology and Evolution*, vol. 10, no. 4, pp. 565–577, 2019.

[66] R. Yacouby and D. Axman, "Probabilistic extension of precision, recall, and f1 score for more thorough evaluation of classification models," in *Proceedings of the first workshop on evaluation and comparison of NLP systems*, 2020, pp. 79–91.



Zhaoqin Wang is an undergraduate student at the School of Information Science and Technology, Dalian Maritime University, Dalian, China, expected to graduate in 2026 and pursue advanced studies. Her current research interests include missing data imputation and related topics in data mining and machine learning.

Qian Ma received her B.S., M.S., and Ph.D. degrees in computer science and technology from Northeastern University, Shenyang, China, in 2011, 2013, and 2019, respectively. She is currently an associate professor at the College of Information Science and Technology, Dalian Maritime University, Dalian, China. Her current research interests include data quality and big data management.

Defu Cui received his B.S. degree in Data Science and Big Data Technology from Dalian Maritime University, Dalian, China, in 2023. He is currently a software development engineer at TravelSky Mobile Technology Co., Ltd., with research interests focusing on missing data imputation, data mining, and machine learning-related topics.

Shikai Guo received his B.S. degree in computer science and technology from Liaoning University, Shenyang, China, in 2011 and the Ph.D. degree in computer applications technology from Dalian Maritime University, Dalian, China, in 2018. He is currently an associate professor with the College of Information Science and Technology, Dalian Maritime University, Dalian, China. His current research interests include intelligent software engineering, EDA software development and optimization, and AI chip design.

Hui Li received his Ph.D. degree in computer architecture from Northeastern University, Shenyang, China, in 2013. He is an associate professor with the School of Information Science and Technology, Dalian Maritime University. His current research interests include intelligent software engineering, mining software repositories, and recommendation systems.

Qiao Ning received her B.S. and Ph.D. degrees from the School of Information Science and Technology, Northeast Normal University, China, in 2019. She is currently an associate professor with the School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi. Her research interests include machine learning and bioinformatics.